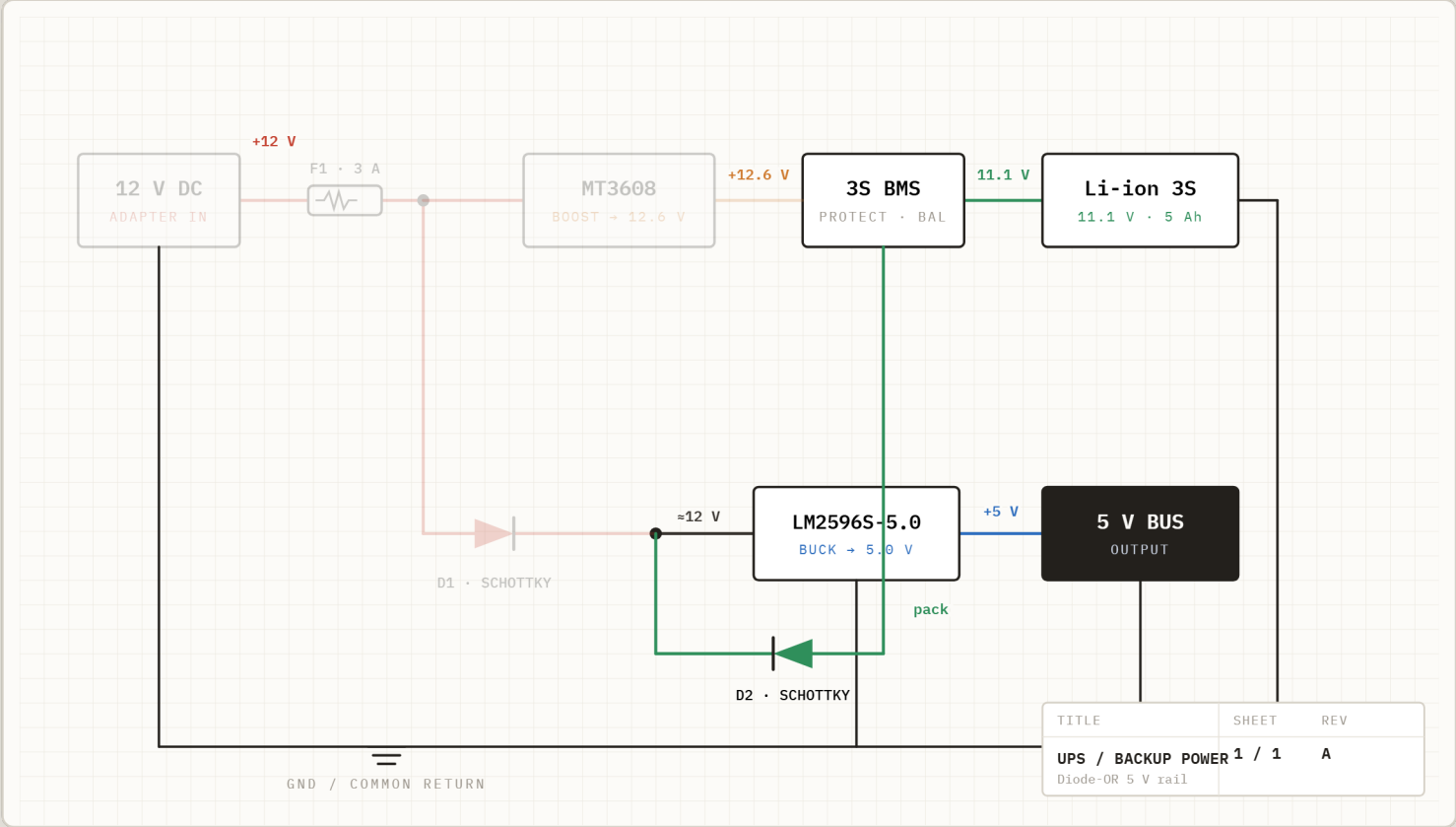


12 V Adapter → 3S Charger + Diode-OR 5 V Backup Bus

OPERATING MODE

MAINS POWER

BATTERY BACKUP



- MAINS LOST – running on battery
- With the adapter gone, D1 reverse-biases and the charger idles. The pack discharges through the BMS and D2 into the OR node, holding the LM2596 input and the 5 V bus alive seamlessly.

RAILS

+12 V in

pack 11.1 V

GND

+12.6 V chg

+5 V out

diode